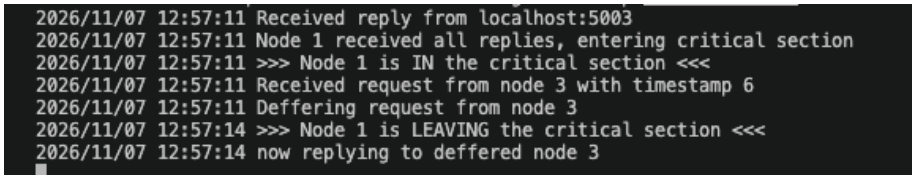
R1.

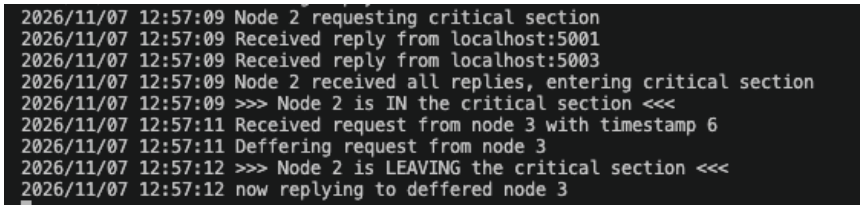
The system has 3 peer nodes, each running in their own port. Every node is both a gRPC server that receives requests and a client which sends requests. The critical section is emulated by a log message and a 3 second sleep. Any node can request access via the command line.

R2.

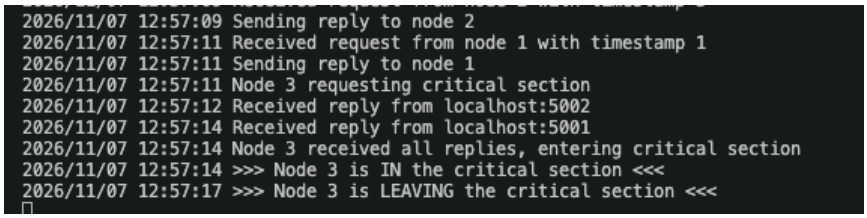
Node 1 terminal:



Node 2 terminal:



Node 3 terminal:



There is no overlap, each node enters critical section after the previous has left. Nodes defer replies when in HELD or when they have priority (Wanted with a lower timestamp)

- Node 1: IN at 12:57:11, OUT at 12:57:14
- Node 2: IN at 12:57:09, OUT at 12:57:12
- Node 3: IN at 12:57:14, OUT at 12:57:17

R3

All 3 nodes eventually entered the critical section, the lamport timestamps ensure ordering with no starvation. Deferred requests are replied to when node exits Critical Section

Algorithm discussion

Node requests -> sets state to WANTED, sends REQUEST to all peers. Receives REQUEST -> reply immediately or defer, depends on the state and/or timestamp. Then it collects all replies and enters critical state (state = HELD). When it then exits CS it replies to all deferred notes

Appendix full terminal logs

Node 1 terminal: (errors here because i was too slow to start the other terminals)

Handin4 % go run main.go --id=1 --port=5001 --peers=localhost:5002,localhost:5003

2026/11/07 12:57:06 Starting node 1 on port 5001 with peers: [localhost:5002 localhost:5003]
2026/11/07 12:57:06 Server listening on port 5001
2026/11/07 12:57:08 Node 1 requesting critical section
2026/11/07 12:57:08 Attempt 1: Request to localhost:5003 failed: rpc error: code = Unavailable desc = connection error: desc = "transport: Error while dialing: dial tcp 127.0.0.1:5003: connect: connection refused"
2026/11/07 12:57:08 Received reply from localhost:5002
2026/11/07 12:57:09 Attempt 2: Request to localhost:5003 failed: rpc error: code = Unavailable desc = connection error: desc = "transport: Error while dialing: dial tcp 127.0.0.1:5003: connect: connection refused"
2026/11/07 12:57:09 Received request from node 2 with timestamp 3
2026/11/07 12:57:09 Sending reply to node 2
2026/11/07 12:57:10 Attempt 3: Request to localhost:5003 failed: rpc error: code = Unavailable desc = connection error: desc = "transport: Error while dialing: dial tcp 127.0.0.1:5003: connect: connection refused"
2026/11/07 12:57:11 Received reply from localhost:5003
2026/11/07 12:57:11 Node 1 received all replies, entering critical section
2026/11/07 12:57:11 >>> Node 1 is IN the critical section <<<
2026/11/07 12:57:11 Received request from node 3 with timestamp 6
2026/11/07 12:57:11 Deffering request from node 3
2026/11/07 12:57:14 >>> Node 1 is LEAVING the critical section <<<
2026/11/07 12:57:14 now replying to deffered node 3

Node 2 terminal:

Handin4 % go run main.go --id=2 --port=5002 --peers=localhost:5001,localhost:5003
2026/11/07 12:57:07 Starting node 2 on port 5002 with peers: [localhost:5001 localhost:5003]
2026/11/07 12:57:07 Server listening on port 5002
2026/11/07 12:57:08 Received request from node 1 with timestamp 1
2026/11/07 12:57:08 Sending reply to node 1
2026/11/07 12:57:09 Node 2 requesting critical section
2026/11/07 12:57:09 Received reply from localhost:5001
2026/11/07 12:57:09 Received reply from localhost:5003
2026/11/07 12:57:09 Node 2 received all replies, entering critical section
2026/11/07 12:57:09 >>> Node 2 is IN the critical section <<<
2026/11/07 12:57:11 Received request from node 3 with timestamp 6
2026/11/07 12:57:11 Deffering request from node 3
2026/11/07 12:57:12 >>> Node 2 is LEAVING the critical section <<<
2026/11/07 12:57:12 now replying to deffered node 3

Node 3 terminal:

Handin4 % go run main.go --id=3 --port=5003 --peers=localhost:5001,localhost:5002
2026/11/07 12:57:09 Starting node 3 on port 5003 with peers: [localhost:5001 localhost:5002]
2026/11/07 12:57:09 Server listening on port 5003
2026/11/07 12:57:09 Received request from node 2 with timestamp 3
2026/11/07 12:57:09 Sending reply to node 2
2026/11/07 12:57:11 Received request from node 1 with timestamp 1
2026/11/07 12:57:11 Sending reply to node 1
2026/11/07 12:57:11 Node 3 requesting critical section
2026/11/07 12:57:12 Received reply from localhost:5002
2026/11/07 12:57:14 Received reply from localhost:5001

2026/11/07 12:57:14 Node 3 received all replies, entering critical section
2026/11/07 12:57:14 >>> Node 3 is IN the critical section <<<
2026/11/07 12:57:17 >>> Node 3 is LEAVING the critical section <<<